$\Rightarrow B = (P^{-1}AP)(P^{-1}AP)$

$= P^{-1}A(PP^{-1})AP$

$= P^{-1}AIAP$

$= P^{-1}A^2P$

Similarly doing in this manner

$B^k = P^{-1}A^kP$ (Assume)

$B^{k+1} = (P^{-1}A^kP)(P^{-1}AP)$

$= P^{-1}A^k(PP^{-1})AP$

$= P^{-1}A^kIAP$

$= P^{-1}A^{k+1}P$

$\therefore B^n = P^{-1}A^nP$

Q:- $f(t) = a_nt^n + a_{n-1}t^{n-1} + \ldots + a_1t + a_0$

$f(A) = a_nA^n + a_{n-1}A^{n-1} + \ldots + a_1A + a_0I$

$f(B) = a_nB^n + a_{n-1}B^{n-1} + \ldots + a_1B + a_0I$

$B = P^{-1}AP$

$f(B) = f(P^{-1}AP)$

$= a_n(P^{-1}AP)^n + a_{n-1}(P^{-1}AP)^{n-1} + \ldots + a_1(P^{-1}AP) + a_0I$

$= a_n(P^{-1}A^nP) + a_{n-1}(P^{-1}A^{n-1}P) + \ldots + a_1(P^{-1}AP) + a_0P^{-1}IP$

$= P^{-1}(a_nA^n)P + P^{-1}(a_{n-1}A^{n-1})P + \ldots + P^{-1}(a_1A)P + P^{-1}a_0IP$

$= P^{-1}(a_nA^n + a_{n-1}A^{n-1} + \ldots + a_1A + a_0I)P$

$= P^{-1}f(A)P$

$\therefore f(A) \approx f(B)$.

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